

Participation User Experience: A Call to Better Manage the Most Important Resource in User-Centered Design

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Abstract

Participating users are the foundation of user-centered design. However, there is a limited understanding of their motivation, engagement, and experience participating in research. In this work, we propose the concept of Participation User Experience (PUX), which addresses participants' experiences in user-centered design. To set a scope for PUX, we conducted a reflexive thematic analysis on workshop data involving 20 experienced user-centered design practitioners and researchers. The analysis yielded five themes, making explicit aspects of PUX that have been implicitly considered and how their consideration could be improved. Great potential lies in addressing intrinsic motivations over extrinsic incentives and developing more structured approaches to planning and measuring PUX to mitigate various sources of bias related to incentives or Experimenter Effects. We contribute to a first understanding of PUX, point to persisting research gaps, and present practical implications for improving participants' experiences in user-centered design.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; **User studies**.

Keywords

User-centered Design, UCD, Experience, Motivation, Engagement, User Experience, UX, Participation User Experience, PUX

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1 Introduction

In Human-Computer Interaction (HCI), and even more so in User-Centered Design (UCD), users and their experiences are at the center of interest. Respective research provides us with an abundance of information on how to engage users with products and design great product experiences [7]. However, to gain such knowledge and create great product experiences, both practitioners and researchers rely on users to participate in their UCD research and help shape the final design outcome [1]. In this paper, we argue that users not only have user experiences (UX) related to the product but also *Participation User Experiences (PUX)*, encompassing the time before, during, and after participation in UCD. Although UCD practitioners and researchers frequently encounter PUX, it is a phenomenon that has not been systematically described, conceptualized, or critically examined. As a result, the little existing knowledge is scattered, and we have no structured overview about what motivates users to participate in UCD, how we can keep them engaged throughout the process, what even makes a *good* experience during participation, and how these factors influence the research outcomes. In doing so, we fail to share and learn how we can navigate and use PUX in UCD processes. To provoke a discourse on PUX and inspire larger-scale investigations, we contribute an initial account and analysis of PUX in UCD processes based on a workshop with experienced UCD practitioners and researchers from industry and academia. With this process-oriented focus, we hope to spark novel conversations that can also help improve methodological rigor.

1.1 What we Know About Participation User Experiences

The following gives a brief overview of PUX aspects that have been discussed so far. It is structured along a triad that is commonly used to conceptualize UX [9, 13]: Before, during, and after the experience. So far, aspects of PUX have mainly been discussed by practitioners, which is why we mainly refer to practitioner contributions in the following.

Like any UX, PUX begins **before participation**, requiring users' awareness and motivation to sign up for UCD research. Users speculate about the process and outcomes, possibly anticipating the

experience. During this pre-participation phase, motivation to participate varies by participant type [30]. Regular users of the product under study may seek product improvement, vent frustrations, or push for change, while panel members, often compensated, may prioritize payment over engagement with the product [3, 30]. Panel members may view participation as a job rather than a meaningful activity [30]. To motivate participation, Stockwell suggests explaining why opinions matter and how they impact design [30]. Another differentiation of motivations to participate is intrinsic (interest-driven) and extrinsic (outcome-driven) motivation [29, 32]. Intrinsic motivators include staying current with technology, sharing perspectives, and social connection [3]. Highlighting the value of user input can engage intrinsic motivation [30, 32]. However, compensation remains the most cited extrinsic motivator [3, 30, 32].

Beyond initial user motivation for participating in UCD, it is vital to identify what sustains a positive experience **during participation**. Practitioners advise offering clear instructions and timely assistance to prevent frustration [3]. In addition, studies should avoid disrupting participants' work [3], adhere to scheduled start and end times [26], and feature concise tasks to reduce fatigue [32]. Privacy must be respected, and personal data handled responsibly, as required by law in many regions [6, 32]. These recommendations – timeliness, transparency, and legal compliance – align with UCD "hygiene factors", ensuring a stable pragmatic experience [34]. In turn, hedonic or even eudaemonic factors to improve PUX have not been explored yet.

Like any UX, PUX can also include **post-participation** experiences – a consideration that UCD research rarely addresses. In other fields, post-participation experiences have been collected to evaluate long-term effects of educational programs [25] or sensitive interviews, e.g., on suicidal thoughts [19]. However, no studies explore this aspect in UCD, opening up research opportunities.

In sum, reports from UCD practice typically examine user motivations for joining or participating but overlook the participation experience itself, post-participation experiences, and the impact of PUX on research outcomes. To facilitate future research and initiate the establishment of guidelines for promoting good participation experiences in UCD, we aim to systematize and enrich discussions on PUX. As a starting point to set the scope of this concept, we conducted a workshop with UCD experts from industry and academia, asking: "What significance does the Participation User Experience have for the UCD process?"

2 Method

2.1 Contributors

The workshop was held as a part of the World Usability Day (WUD) 2023 initiated by the German Usability and User Experience Professionals Association to link research and industry and foster exchange. We issued an open call on the WUD website, mailing lists, and social networks, and directly invited industry and academic contacts, attracting 20 participants from diverse fields in academia ($n = 5$) and industry ($n = 15$). Participants, referred to as *contributors*, brought expertise from various domains, with some contributing expertise from multiple domains. The domains included healthcare ($n = 6$), air traffic control ($n = 2$), aerospace ($n = 1$), road traffic ($n = 1$), consumer product design ($n = 5$), web

services ($n = 6$), inclusive design ($n = 4$), civic participation ($n = 3$), education ($n = 1$) and participatory design ($n = 5$). The contributors' educational backgrounds included HCI ($n = 14$), Psychology ($n = 3$), Computer Science ($n = 1$), Design ($n = 1$), and UX ($n = 1$). They worked in UX departments of large companies or government institutions ($n = 9$), academia ($n = 5$), start-ups ($n = 3$), or as UX agency professionals or freelancers ($n = 3$). At least those trained in HCI and Psychology ($n = 17$) had also been participants themselves, offering insights from both expert and user perspectives. To protect the contributors' privacy, no further demographic data was collected.

2.2 Workshop Planning, Setting & Procedure

Over three months, authors MH and SH used asynchronous brain-writing inspired by the collective notebook [33]. Author SH identified the research gap around PUX, listing suspected factors and questions. Author MH explored scientific literature and other sources for inspiration. Regular meetings helped refine questions and optimize the workshop concept. Simultaneously, a call for participation invited contributors to submit abstracts on their touchpoints and questions about PUX. Based on these submissions, the workshop concept was finalized and structured around six core questions: (1) What does your typical UX process look like? (2) What (de)motivates our users today? (3) How do we deal with different motivations? (4) What is good Participation UX? (5) What unwanted effects can "too good" of a PUX trigger? (6) What does a PUX-centered research process look like?

The workshop was held in a local design thinking lab with flexible spaces for group work. After a general introduction to the workshop, contributors introduced themselves. Contributors formed five groups (A to E) of three to four members, evenly mixing academics and practitioners. Groups worked on one question at a time for 20 minutes, documenting ideas on flip charts. After three questions, a plenary session allowed groups to share insights, followed by a break for refreshments and broader discussion. The workshop concluded with a final plenary session where groups shared takeaways, which were not recorded for analysis.

2.3 Analytical Process

To analyze the data, all analog content was digitized on a Miro whiteboard (RealtimeBoard Inc, Delaware, USA). After initial familiarization in Miro, the data was uploaded to MAXQDA (version 24.2.0, Verbi GmbH, Berlin, Germany) for qualitative analysis.

Using Braun and Clarke's reflexive thematic analysis [4], the analysis process was primarily inductive, with some deductive structuring based on workshop questions. Author MH could draw on her background as both a UCD participant and study conductor, allowing her to include a UCD expert's and participant's perspective while conducting the analysis. The dataset was coded once, refined by renaming or merging similar codes, and re-coded for thoroughness. Initial theme generation produced seven candidate themes with subthemes linked to enhancing participant experiences. After consulting author SH, author MH refined the themes, reducing them to five coherent themes relevant to answering the research question. Finally, the themes were named, defined, and written up

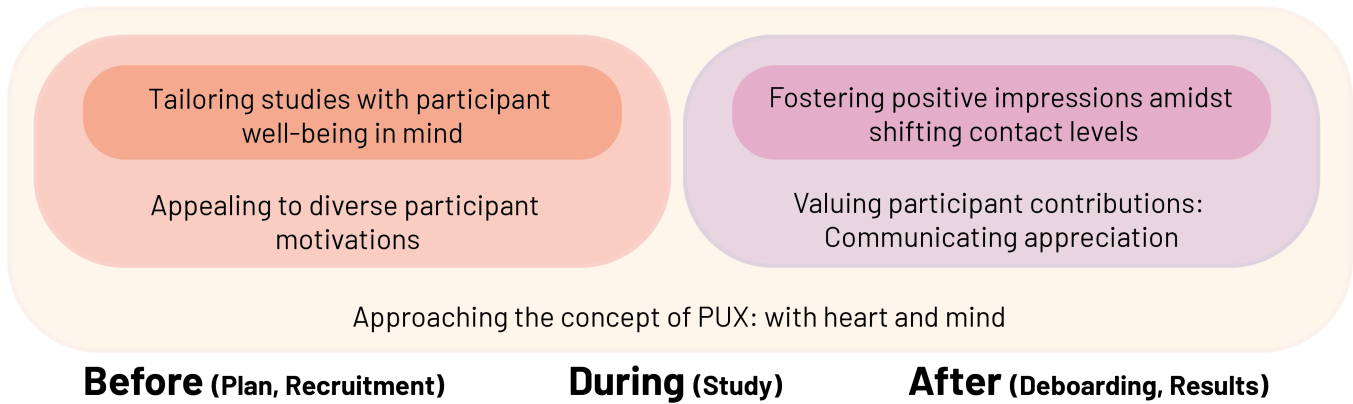


Figure 1: The five themes, their interrelationships depicted as surrounding bubbles, and their main significance during the PUX triad: before, during, and after participation.

as will be presented in this paper. Selected statements, analyzed in German, were translated into English for reference in the paper.

3 Analysis

The following analysis makes explicit common implicit considerations of PUX in UCD processes and suggests ways to improve their integration. All five themes are illustrated in Figure 1 and the Appendix A. Given that the themes *Appealing to diverse participant motivations* and *Valuing participant contributions: Communicating appreciation* share significant overlaps with previous work (Section 1), we will focus on the other themes in what follows. Each theme is introduced with a general description of its contents, followed by several subthemes detailing the theme, each with *italic* titles. Additionally, we draw comparisons between our findings and previous work and highlight considerations for practice and future work.

3.1 Tailoring Studies With Participant Well-Being in Mind

The first theme highlights how participants' well-being is considered before and during participants' participation in UCD processes to generate a good PUX (Figure 1). It is influenced by the theme *Appealing to diverse participant motivations* in that diverse motivations might require diverse considerations in relation to well-being.

Contributors highlighted recognizing participants' individuality to create settings that allow people to bring their authentic selves to the study. They even considered *customizing studies to participating users' needs*. So far, this is mainly discussed concerning accessibility and how to make UCD accessible [e.g., 5, 12, 17]. Planning studies to include people with disabilities requires customized considerations. However, despite the common practice of considering user needs in product design more generally [e.g., 18], such considerations seldomly transfer to the study design. Thus, in practice, PUX could be improved by more nuanced considerations of participating users' needs in UCD planning.

Contributors also highlighted that the type of study influences PUX consideration during planning. They considered *adaptable planning for different study types*, including different UCD methods (e.g., standardized lab studies, unsupervised home prototype

tests, "user test of the current feature" [Group D]), project stages, or study contexts. Due to the diversity of UCD study types [e.g., 21], study planning and, in turn, PUX planning is diverse, too. Therefore, creating good PUX may not have a one-size-fits-all solution for UCD studies, and in practice, each study requires individual consideration. That individual consideration and whether certain study types may be particularly suitable (or unsuitable) for good PUX poses a topic for future research.

A third consideration raised by contributors was *planning for participant comfort*. This includes sufficient breaks and a reasonable study length, but also drinks or snacks to "create a good atmosphere" (Group D). While an adequate task length echoes previous considerations [32], study atmospheres have not been discussed for UCD studies. Additionally, almost all groups (A, B, C, E) mentioned "fun" as an essential positive part of study design but pointed to some concerns related to too much fun, for example, "Validity of the answers[,] Time[,] Costs" (Group E). Thus, in practice, PUX could be improved by ensuring participant comfort through breaks, adequate length, a welcoming atmosphere, and even fun. However, future work needs to determine the impact of participant comfort and specific mechanisms, such as fun, on study results.

A last consideration was to actively prevent negative experiences for participants and *alleviate participant pitfalls*. For example, the study should not be "boring" (Groups B, C, E), too exhausting, or easily lead to frustration. Avoiding frustration and stress is something Boyd already pointed out [3], highlighting its importance. In the eyes of our contributors, another negative experience is to feel being examined, for instance, if too many observers are present. A perceived "lack of time" (Group B) was also considered harmful to PUX. Thus, in practice, PUX could be improved by minimizing boredom, exhaustion, and frustration while reducing feelings of being examined or rushed. For example, contributors mentioned that user testing should be done with prototypes that are "test-ready", meaning that regardless of their fidelity, they should work smoothly (or be operated smoothly in wizard-of-oz scenarios) to enable meaningful insights and avoid frustration due to prototype failures.

3.2 Fostering Positive Impressions Amidst Shifting Contact Levels

Another aspect influencing PUX is the impression the study conductor makes on the participants through communication and demeanor. However, not all interactions are in person, and consciously shaping the few shared interactions is especially important. The theme is influenced by the theme *Valuing participant contributions: Communicating appreciation*, as an appreciative attitude is a basic prerequisite for conveying positive impressions.

Not everyone has an opportunity for personal interaction with their participants, as *contextual factors influence user touchpoints*. This is also reflected by the growing body of literature on remote research (see [20] for an overview). Yet, especially personal contact allows study conductors to impact and even improve PUX for participating users. Various factors influence if and how study staff will be in contact with participating users and thus have the possibility to shape PUX through contact. Sometimes, communication with participants is handled by “(Company) internal intermediaries, e.g., sales + customer support” (Group B), or recruitment is handled through “external agencies for user acquisition” (Group E). Other times, recruitment is based on personal connections, resulting in high amounts of user touchpoints. The amount of face-to-face contact with participating users can vary greatly and is hard to generalize. Depending on the setting, PUX might be influenced not only by the study conductors but also by third-party interactions with participants. Thus, in practice, planning PUX could require the inclusion of more than the immediate research team, and it becomes important how others are instructed about PUX, too.

Limited interactions with the participants increase the relative importance and impact of the study conductors’ communication on their experience. As a result, every word counts, and our contributors highlighted that one should try *achieving positive impact through precise wording*. Boyd partly mentioned this, stating that study instructions need to be clear [3]. While there is an awareness of the importance of language in scientific communication [11] and how language influences a person’s thinking [14], to our knowledge, there is no literature leveraging these findings specifically for enhancing participants’ experiences in UCD. Our contributors pointed out that the way study conductors present their study – the “framing” as Group E puts it – is vital for shaping first impressions about the study and that good reputations of the organization could improve PUX since participants might be more motivated to attend. Thus, in practice, PUX could be improved by careful wording of the study announcements, materials, and instructions. For example, contributors repeatedly mentioned the importance of using the language of the users in these materials rather than domain-specific terms that might confuse participants.

Another consideration for PUX is to *create positive experiences through transparent communication*. Keeping the participants informed at each step of the way can help them feel comfortable and know what to expect. Group A captures this sentiment: “Know what it is about / what the goal is / what happens with [the] results”. From the perspective of our contributors, transparent communication also includes restating the value of users’ participation. Once a study is completed, participants may be interested in the results. Stockwell notes that especially long-term users appreciate knowing

what was changed based on their feedback [30]. Thus, in practice, PUX could be improved by transparent communication practices that help participants manage expectations and understand the value of their participation, as well as how they contributed to the final outcome.

Lastly, study conductors’ presentations can also influence PUX. Contributors emphasized the importance of being friendly, motivated, competent, and empathetic towards participants, showing high demands and responsibilities. Yet, *study conductors’ demeanor shapes participant experience*, and how participants are approached can shape their experience for better – or worse. This poses some parallels to the Experimenter Effect first described by the psychologists Rosenthal and Fode as the biasing influence of the behavior or other cues sent by the experimenter on participants’ performance [28]. Generalizing Rosenthal and Fode’s observations for UCD studies, the assumption is that the participants’ experience moderates the influence of the study conductor on participants’ responses. The extent of this effect remains a topic for future research. Thus, in practice, study conductors should adopt a professional but friendly and empathetic demeanor to improve PUX and minimize unintentional bias.

3.3 Approaching the Concept of PUX: With Heart and Mind

While the previous themes discussed how various aspects across UCD processes influence PUX, the theme *Approaching the concept of PUX: With heart and mind* highlights how more than individual aspects are needed to address PUX in UCD processes: a fundamental PUX attitude. Consequently, the theme influences all other themes.

Touching onto the “mind” part of the theme, many contributors wished for a “Guideline” (Group D) or “Checklist” (Group E) to plan PUX. Up to now, PUX has been the implicit, unguided responsibility of respective study conductors and thus, our contributors wish for *planning for PUX consistency*. The contributors seem to agree that – similar to a pleasant UX – to create good PUX, study conductors need to plan it but also follow it up with evaluations. Thus, an essential area for future research is how to integrate PUX planning and evaluation in existing UCD processes.

To create good PUX and keep it consistent throughout or across several studies, it is essential to reflect on PUX regularly. For this reflection, our contributors wished for *simplicity in PUX assessment*. Contributors considered different ways of capturing PUX and concluded that asking participants about their PUX may prove to be the best way, while a questionnaire might be unfitting. Other ideas include looking at dropout rates, which indicate a bad experience but offer no cues on the underlying reasons for the study cancellation. Thus, in practice, study conductors could consider evaluating PUX aspects as part of pilot studies and, later on, inquiring participants about their PUX. The extent and form of this inquiry is a question for future research.

Apart from concrete guidelines and measurements, there was also the notion that *developing a feel for PUX* is essential. As a precondition for PUX measurements and interventions, researchers in UCD need to develop an “awareness for PUX” (Group E). Getting a sense of PUX also calls for a change of perspective, looking at things from the participant’s point of view. Some contributors even

said that PUX is “a matter of feeling” (Group D). Instigating this change of perspective is the aim of this publication and the first important step for future consideration of PUX.

Planning, reviewing, and optimizing PUX can feel like a whole “research process for a research process” (Group D). This is why our contributors considered *balancing effort with outcome* to not put an excessive amount of effort into PUX. After all, tailored communication (section 3.2) or extra planning can easily become too much to handle, aside from financing and time needed. As Group D put it, “accepting ‘Good enough solutions’” can be important, although due to the current disregard of this topic, what constitutes a “good enough” experience for participants is unclear and remains an important question for future work.

As a final caveat, on top of the effort associated with measuring and shaping PUX, even good things can be overdone. In this case, too much PUX may lead to a distortion of the data. Thus, study conductors need to balance PUX and *find a sweet spot in avoiding bias*. For example, providing a too-pleasant atmosphere may lead to overly positive answers due to the Halo Effect [31], which Stockwell linked to preexisting impressions of a company or product in the context of UCD [30]. Our findings suggest that this effect may also extend to the process itself. Similarly, an overly friendly study conductor can lead to a “social desirability bias” (Group A), first described by Edwards as participants giving more socially desirable answers [8], e.g. in UCD conforming to the study conductors’ opinion. Further sources of bias, like pity or a novelty effect, may lead to “too benevolent answers/ no criticism” (Group E). Nonetheless, in practice, a good PUX should still be sought, as participants may be more inclined to share serious criticisms if they do not simply rush the study or believe their answers will have no impact, as highlighted by the previous themes.

4 General Discussion

This research aimed to explore PUX and its impact on UCD processes, practices, and outcomes at different stages – before, during, or after participation. Insights from our UCD expert workshop make explicit existing practices and considerations for improvement and future research. User participation, central to UCD in both academic and industrial settings, informs design decisions published as papers or products. Hence, understanding participation experiences and their effects on research outcomes is crucial for action, intervention, and iterative learning. We need to understand PUX (i.e., theoretical implications) to better plan and decide how to act (i.e., practical implications).

4.1 Theoretical and Practical Implications and Future Work

The workshop highlighted that UCD practitioners recognize participants’ diverse motivations [3, 30, 32] despite limited research on PUX. However, addressing motivations depends heavily on the individual study conductor, introducing subjective biases that remain unexamined. To provide a more hands-on approach to *developing a feel for PUX* and support UCD novices or those new to a domain, future research should develop guides for acting on diverse participant motivations and enabling empathetic, tailored interactions. A practical consideration for future work is to consider dedicated

PUX budgets. Even study conductors with the communicative skills for handling participants with differing motivations cannot make up for organizational shortcomings, for example, regarding the provision of snacks and drinks.

Besides motivational factors, UCD researchers and practitioners must address biases arising during the process. Biasing effects from researchers or setups, noted since the 1930s [16, 22, 27], remain underexplored in UCD’s varied settings, from online click tests to long-term, in-context studies. How social and motivational biases influence outcomes across these scenarios requires further study.

First, our analysis supports the notion that the Experimenter Effect [28] could be moderated by the participants’ study experience, warranting further research within UCD. Similarly, our findings imply that Edwards’ Social Desirability bias [8] appears relevant to UCD, especially concerning study conductors. Future research should explore how PUX affects these biases and identify methods to reduce them. Striking a balance in PUX is crucial; higher PUX levels risk increased bias, but a certain level seems necessary for genuine feedback.

Second, our analysis suggests that participants may have a particularly positive or negative experience throughout the study, leading to a biased opinion about the product at hand, possibly due to the Halo Effect [31]. While previously linked to products or brands [24, 30], the link to UCD processes is a novel perspective that requires investigation.

Third, our analysis suggests that incentives, particularly monetary, introduce bias, echoing Hsieh and Kocielnik’s findings for applied research [15]. Hence, future research should explore strategies to increase intrinsic motivation. First clues can be found in the works of the UXtweak Team, Stockwell, and Boyd [3, 30, 32], which suggest highlighting participant value, explaining why their opinion is useful and how it influences the process, focusing on altruistic motives, and promoting social interactions to foster intrinsic motivation. Our work echoes these strategies and adds that interest, too, is of importance for participants’ intrinsic motivation. However, intrinsic motivation alone cannot suffice; fair compensation (e.g., monetary or through an event [10]) remains vital to ensure ethical participant treatment.

In general, the current approach to PUX often begins with extrinsic motivators like monetary incentives, addressing intrinsic motivators such as social connection or curiosity only later [3]. Our analysis suggests that great potential lies in considering intrinsic motivations from the start. Study conductors can achieve this by valuing participants, communicating this appreciation clearly, and providing different ways to create value for participants. As a caveat, participants with more autonomous levels of motivation systematically report higher needs for satisfaction [2] – an effect biasing the interpretation of quantitative UX measures. To create value, it is essential to use structured, transparent communication. Keeping participants informed, comfortable, and prepared is essential [34], along with considerate scheduling [3, 26], respecting privacy [32], and measuring their initial motivation to explain true effects of the product.

Further, our analysis provides first ideas on how to foster eudaimonic experiences to create value and meaning in PUX. Applying Mekler and Hornbæk’s framework of meaning [23] to PUX, an awareness for *connectedness* to one’s worldview is what we have

been calling for repeatedly throughout this paper: recognizing participants' individuality and how this impacts motivation, experience, and meaning-making is essential. *Purpose* and *significance* can be emphasized by showing participants the value of their input [30, 32] and informing them of changes based on their feedback [30]. Clear communication and expectation management enhance *coherence*, ensuring experiences align with expectations. *Resonance*, whether an experience feels "right", appears to be a factor one can hardly influence and thus neither our analysis nor the mentioned literature provide any insights on how it could be improved.

Lastly, creating hedonic experiences for participants remains unexplored in UCD. After all, participating users, too, are people who strive for hedonic experiences such as stimulation. So far, it is unknown if hedonism is a suitable motivator for participation, e.g., by providing particularly interesting use cases. Optimizing studies for hedonism may attract certain users while alienating others (e.g., more eudaemonically motivated participants), and it might increase biasing effects, especially the Halo Effect.

Overall, it is time to adopt the user-centered focus to a meta-level that includes the study design – or, in other words, to focus on PUX in UCD projects. For this purpose, we provide a first conceptualization of PUX and uncover directions for future work. Ultimately, with a turn to process-oriented research, both researchers and practitioners can improve their methodological rigor by being more aware of different sources of bias while providing a better experience for their participating users.

4.2 Limitations

A limitation of our study is the choice to invite researchers and practitioners as contributors. We decided on this sample based on our goal to initiate a discourse on PUX, explore how it might (already) impact UCD practices and set the scope for larger-scale investigations in the future. Contributors with decades of user research experience shared valuable insights across diverse user groups, domains, and contexts. Many had also participated in UCD studies (e.g., during education), which also brought in the participant's perspective. Nonetheless, future work must explore current UCD participants' views across domains and methods to identify what defines a pleasant PUX and how factors like study setup, personnel, procedure, or user group shape it.

A second limitation concerns the analytical approach. Reflexive thematic analysis suited the diverse workshop data created by the contributors. However, the contributed data points frequently consisted of rather short statements, which complicated understanding the context and cohesion. This allowed for more interpretative leeway, possibly diverting from the original meaning, but also opened the possibility of looking at the data from many perspectives, for instance, relating the domain of the data. Since the lead analyst also participated in the workshop, some context was retained.

5 Conclusion

We introduced the new term *Participation User Experience* – short PUX – to provide a label for research about the experience of participating users in UCD. Making a previously implicit and under-researched concept explicit, we strive to advance its consideration

in scientific research and practical application. Through the qualitative analysis of workshop data centered around the concept of PUX, we make explicit how PUX has implicitly been considered in diverse fields in both academia and industry and give first directions for future research on this topic. In short, we underline the need for more research around participating users' motivations and experiences, as well as the possible biases introduced by the current lack of considerations of PUX. The current practice of entrusting the empathetic abilities of the study conductor to create some form of PUX is not enough when considering the significance of design decisions built upon this trusted user data. One practical path to close the issue of lacking PUX consistency lies within a systematic and comprehensible approach to planning, providing, and measuring PUX. Here lies the most challenging angle for future research on PUX: to identify (individual) factors speaking to participating users' pragmatic, hedonic, or eudaemonic motives in a way that ensures their continuous participation throughout the process without a spill-over impact on the product UX.

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A Summary of All Themes and Subthemes

Table 1: Overview of the theme *Tailoring studies with participant well-being in mind* and its subthemes. The theme particularly concerns PUX before and during participation.

Tailoring studies with participant well-being in mind	
This theme discusses the planning phase of a user study and details how there are many variables influencing PUX that need consideration at this stage. Keeping factors like participant well-being in mind during the planning phase allows for a consistent PUX for all participants and reduces the workload for the study conductors later. Different from the next theme focusing on motivating participants to join the study, this theme highlights how planning for participant well-being ensures positive PUX throughout the study process.	
Customizing studies to participants' needs	Each participant group (e.g., people with disabilities, medical professionals) has distinct needs that should be accommodated in study design.
Adaptable planning for different study types	The type of user test (e.g., lab-based, unsupervised) influences planning and study setup itself.
Planning for participant comfort	Consider participant well-being by incorporating breaks, reasonable study length, and fostering a positive atmosphere.
Alleviating participant pitfalls	Avoid negative elements such as boredom, frustration, exhaustion, excessive observation, and time constraints in study design.

Table 2: Overview of the theme *Appealing to diverse participant motivations* and its subthemes. The theme particularly concerns PUX before and during participation.

Appealing to diverse participant motivations	
This theme offers an insight into the factors leading to participation in a user study, discussing the diverse motivations of participants in UX studies from the viewpoint of those conducting the studies. While not an exhaustive list, it showcases different aspects to consider in advertising user studies and therefore improving the PUX early on, as well as motivational factors important to improve PUX during the study period. The theme further highlights how motivations should be addressed early on, even during recruitment and initial contact.	
The appeal of novelty	Participants are motivated by the chance to experience something new, like exploring new places or interacting with novel technology.
Broadening the knowledge horizon	Participants are motivated by learning opportunities, such as gaining new skills or insights related to their work or self.
Building bonds, not just studies	Social interaction, such as exchanging ideas and building relationships with study staff, can motivate participants.
Passion sparks intrinsic motivation	Intrinsic motivation, driven by genuine interest in the study's subject or personal benefit, is important for sustaining engagement.
Finding purpose in participation	Some participants are motivated by the belief that their participation is meaningful and contributes to the greater good.

Table 3: Overview of the theme *Fostering positive impressions amidst shifting contact levels* and its subthemes. The theme particularly concerns PUX during and after participation.

Fostering positive impressions amidst shifting contact levels	
This theme discusses the importance of the impression a study conductor makes on the participants through the way they communicate and present themselves. It seems that the attitude of the study conductor is crucial for the participants' comfort. However, the theme also points out how it might not always be possible to interact personally with the participants, which makes it difficult to influence this portrayed image. This is where shaping the few shared interactions are especially important to create a good PUX for participants. Unlike the previous theme, which considers the study design, this theme focuses on interpersonal factors mainly during the participation in UCD.	
Contextual factors influencing user touchpoints	Participant communication varies depending on factors like third-party intermediaries or recruitment methods, possibly preventing direct contact with the researcher.
Achieving positive impact through precise wording	Careful and precise communication, including study framing and organization reputation, can shape participant impressions.
Creating positive experiences through transparent communication	Keeping participants informed about the study's goals, process, and results builds trust and positive experiences.
Study conductor's demeanor shapes participant experience	The attitude of the study conductor (e.g., being friendly, motivated, and empathetic) impacts participants' overall experience.

Table 4: Overview of the theme *Valuing participant contributions: Communicating appreciation* and its subthemes. The theme particularly concerns PUX during and after participation.

Valuing participant contributions: Communicating appreciation	
This theme highlights the mindset that every participant is valuable to the studies' goals, and how it is especially important to convey this appreciation to the participants. By acknowledging the value of participants' contributions and communicating it appropriately, participants better understand their role in the user study, therefore improving their PUX. Unlike the previous theme that focused on interaction in general, this theme specifies how researchers' attitudes toward participants influence their sense of contribution and value during the UCD process.	
Valuing users throughout the process	Participants' contributions are valuable at all stages, from early involvement to feedback during product iterations, ideally with long-term engagement.
Respecting participants and their contributions	Treating participants with respect, valuing their opinions, ensuring informed consent, and showing appreciation, e.g., through appropriate compensation, are essential
Inadequacy of current compensations	Current incentives, especially monetary ones, may not always be effective or ideal, e.g., relating to issues with motivation.
Emphasizing the value of participation	It's crucial to manage expectations and demonstrate the significance of participants' contributions to make them feel valued and impactful.

Table 5: Overview of the theme *Approaching the concept of PUX: With heart and mind* and its subthemes. The theme concerns PUX at all stages of participation (before, during, after).

Approaching the concept of PUX: With heart and mind	
With the previous themes discussing PUX at specific study phases, this theme focuses on more fundamental aspects of PUX such as a general attitude and practice that influences PUX across all phases of a UCD process. It shows how both a strategic approach and a general sense for good PUX is needed in order to best integrate it in practices.	
Planning for PUX consistency	Consistent PUX requires careful planning, regular reflection, and evaluation across studies to ensure a good PUX.
Simplicity in PUX assessment	PUX measurement should be simple and minimally burdensome, e.g., through direct participant feedback.
Developing a feel for PUX	Beyond formal guidelines, creating an understanding and awareness of the participant's perspective, is essential.
Balancing effort with outcome	Efforts invested in enhancing PUX must be proportional to the outcomes, and "good enough solutions" may sometimes suffice.
Finding a sweet spot in avoiding bias	Overly positive atmospheres or overly friendly conductors can create biases, so it's crucial to balance PUX to avoid distorted data.